Electrocution / Electrical Injuries

History

- **Power source** – lightning vs. electrical
- **Voltage** – high (industrial / power lines) vs. low (household)
- **Current** – alternating (household) vs. direct (industrial / power lines)
- Did patient have tetany and hold onto power source for prolonged period of time?
- Did the patient fall? Was the patient thrown to the ground?

Physical

You must do a thorough physical exam on these patients. Undress them completely to look for burn, entrance / exit wounds, or other trauma.

- Vital Signs
- EKG – Cardiac arrhythmias are very common and patient should be closely monitored
- Respiratory – These patients are at high risk for respiratory arrest, chest wall tetany, and pneumothorax
- Skin – Burns may range from mild-appearing (these can be deceiving as to internal injury) to severe. Look for oral burns on children from electrical cords.
- Mental status / Neuro – seizures, altered mental status, amnesia, paralysis
- Other signs of trauma – spinal, musculoskeletal
  - High risk for spine injury

Notes

- **Low voltage injuries** – may cause spectrum of problems from minor burns to cardiac arrest
- **High voltage injuries** – cause severe burns, internal injury, more likely to cause blunt trauma (from falls / being thrown)
- **Lightning injuries** – more likely to cause neurologic injury. In a mass casualty incident from a lightning strike -- **treat the patients in cardiac arrest first.** These patients have a high likelihood of survival.

EMT/AEMT

- **Treat cardiac arrest victims first**
- Follow General Medical Care Guideline
- Follow Adult Resuscitation Guideline, as indicated
- Airway management as indicated
- Control hemorrhage

Paramedic

- Advanced airway as indicated
- Cardiac monitoring
- Pain control per Pain Management Guideline
- Treat seizures per Seizure Guideline
- Transport as Burn patient per Burn Guideline